

Jiyang Lee

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Research Interests

Robot Learning, Embodied AI, Vision-Language-Action Models, World-Action Models

Research Experience

University of Waterloo / Seoul National University

Jan 2026 – Present

Generative Robot Policies on Real Robots

- Fine-tuned and evaluated $\pi_{0.5}$ (full and LoRA), Diffusion Policy, and OpenVLA in LIBERO and on a Kinova Gen3 robot.
- Built an end-to-end real-robot learning pipeline for demonstration collection, policy training, and closed-loop deployment.
- Conducted controlled studies of generative policies in multi-object manipulation settings.
- Implemented inference-time policy guidance using learned visual dynamics.

Seoul National University

Sep 2026 – Present

Humanoid Data Generation and Whole-Body Tracking

- Processed G1-retargeted motion data from OMOMO and BONES-SEED, selected and annotated eight indoor 3D Gaussian Splatting scenes, and built an egocentric rendering environment in Isaac Sim.
- Checked retargeted motions in MuJoCo using a public whole-body tracking controller.
- Designed a lower-body intervention diagnostic set (27 states $\times$ 8 seeds) for later execution analysis.

Education

Seoul National University, Seoul, Korea

Mar 2024 – Present

Ph.D. Candidate in Civil, Urban and Environmental Engineering

Advisor: Prof. Seong-Woo Kim

Expected graduation: August 2028

University of Waterloo, Waterloo, Canada

Jan 2026 – Jul 2026

International Visiting Graduate Student, Electrical and Computer Engineering

Advisor: Prof. Roya Firoozi

Sogang University, Seoul, Korea

Mar 2015 – Feb 2017

Master of Science in Mechanical Engineering

Advisor: Prof. Cheol-soo Lee

Graduate Research Assistant, CAD/CAM Lab (03/2015–02/2017)

Sogang University, Seoul, Korea

Mar 2011 – Feb 2015

Bachelor of Science in Mechanical Engineering

Undergraduate Research Assistant, CAD/CAM Lab (12/2013–02/2015)

Professional Experience

LG Display Co., Ltd., Paju, Korea

Dec 2020 – Feb 2024

Research Engineer, Vision Algorithm Team

- Developed a few-shot classification model for OLED defect images.
- Built an AI-based automatic anomaly detection and alarm system.
- Accelerated legacy rule-based anomaly system with CUDA, improving speed by 7 times.

Korea Institute of Science and Technology (KIST), Seoul, Korea

Oct 2018 – Feb 2020

Robotics Researcher

- Designed and assembled a quadruped robot.
- Developed a panoramic-view system for a firefighting armored robot.
- Conducted UWB-based robot swarm localization experiments.

- Built vision and XRF inspection machines and OLED line loading equipment.

Publications

Under Review

- **Jiyang Lee**, Roya Firoozi, and Seong-Woo Kim, "Blind review," submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.

Journals

- **Lee, J.Y.** and Lee, C.S. (2018). *Path planning for SCARA robot based on marker detection using feature extraction and labelling*. International Journal of Computer Integrated Manufacturing, 31(8), 769-776.
- **Lee, J.Y.**, Lee, D.G., and Lee, C.S. (2017). *Development of Robotic Dispensing System for Radiopharmaceuticals using SCARA Robot*. Journal of the Korean Society of Manufacturing Technology Engineers, 26(5), 441-449. [Best Paper Award]

Conferences

- Chan Kim, Keonwoo Kim, Mintaek Oh, Hanbi Baek, **Jiyang Lee**, et al., "E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models," IEEE ICRA, Atlanta, USA, May 2025.
- Kim, M., **Lee, J.Y.**, et al., "A Robust Position Estimation Algorithm under Unusual Large Range Errors," IEEE ROMAN, New Delhi, India, 2019.
- **Lee, J.Y.** and Lee, C.S., "Path planning for SCARA robot based on marker detection using feature extraction, labeling, and inverse perspective transform" 26th International Conference on Flexible Automation and Intelligent Manufacturing, Seoul, 2016.

Awards

- Best Paper Award, Korean Society of Manufacturing Technology Engineers, 2017

Additional Information

- **Robot Learning:** $\pi_{0.5}$ (openpi), Diffusion Policy, OpenVLA, LIBERO, MimicGen, Isaac Sim, MuJoCo
- **Programming / ML:** Python, C++, PyTorch, OpenCV, CUDA
- **Mechanical Design:** Autodesk AutoCAD, Inventor, SolidWorks